Q.4. (a): Using the semi-classical model, show that the electron moves on a surface of constant energy in k-space. (5)

(b): Define and derive the expression for the effective mass of an electron. (5)

UNIVERSITY OF THE PUNJAB

BS (After Associate Degree) : Sixth Semester – Spring 2023

Roll No. [redacted]

Paper: Solid State Physics-II   Course Code: PHYS-3402   Time: 3 Hrs.   Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answer of the following questions: (15x2=30)

1. Define fermi energy of metals. How does it depends on concertation of electron in metals of the sample?
2. Give the schematic representation of variation of f(E) with energy of metals at T=0K, 300K and infinity
3. Discuss the variation in conductivity of metals with rise in temperature and impurity content.
4. Comment on conductivity when a band is filled (i) more than half (ii) completely.
5. How you will identify N-type or p-type semiconductor.
6. Write down the some properties of holes.
7. State the bloch theorem. Discuss its importance in band theory.
8. Compare the results of heat capacity of electron gas calculated classical theory and quantum theory.
9. Is there any possibility of holes conduction in metals? If so justify.
10. How the free electron theory does explains the observed small values of specific heat of metals?
11. What do you understand by fermi gas?
12. Give the condition of crystal reflection and discuss its consequences.
13. Explain the impact of temperature on mobility of semiconductor.
14. Explain the principle of Hall's effect with the help of relevant diagram.
15. Will the impurities in semiconductor effect the conductivity of semiconductor?

Answer the following questions.

Q2. What is meant by effective mass of an electron? What is its significance? Show that mass of electron in crystal is inversely proportional to the second derivative of E vs k curve. Using this curve calculate the effective mass of electron at different points. Discuss the conditions when the effective mass of an electron becomes positive, negative and infinite. (10)

Q3.
(a) Distinguish between a metal, insulator and semiconductor on the basis of energy band structure.
(b) What are symmetric and antisymmetric wave functions of electron? Explain how they are the reasons of producing band gap in solids. (5+5)

Q4.
(a) Describe Hall's effect. Explain how the measurement of the hall's coefficient helps one to determine the mobility of electron in metals.
(b) State law of mass action. Give its significance. (6+4)

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SSP-II

Past Paper Spring-2023

Short Questions:

1 Define fermi energy of metals. How does it depends on concertation of electron in metals of the sample?

Fermi energy is the highest energy level occupied by electrons in a metal at absolute zero temperature (0 K).

→ It represents the energy of the most energetic in a metal when all lower energy states or filled.

OR

"The fermienergy E_f is defined as the energy of the topmost filled level in ground state of N-electron system."

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$$\boxed{E_f = \frac{\hbar^2}{2m}\left(\frac{N\pi}{2L}\right)^2}$$

Dependence of Electron Concentration:-

Fermi energy E_f depends on the electron concentration n is a metal as:

$$E_f = \frac{\hbar^2}{2m}\left(3\pi^2 n\right)^{2/3}$$

where,

- $\hbar$ is the reduced Planck's constant.
- m is the mass of an electron
- n is the number of conduction electrons per unit volume.

→ The Fermi energy is directly proportional to the $\frac{2}{3}$ power of the electron concentration.

That means:-

→ If the electron concentration increases the Fermi energy increases.

→ Metals with higher free electron density will have higher Fermi energy.

Question # 2

Give the schematic representation of variation of F(E) with energy of metals at T 0k, 300k and infinity?

Answer:-

The Fermi-Dirac distribution function f(E) gives the probability that an energy states at energy E is occupied by an electron.

$$f(E) = \frac{1}{1+e^{(E-E_f)/kT}}$$

- E_f is the fermi energy
- k is the Boltzman constant.
- T is the absolute Temperature.

Behavior at Different Temperatures:-

1. At T = 0

- $f(E) = 1$ for $E < E_f$
- $f(E) = 0$ for $E > E_f$
- sharp step function at $E = E_f$

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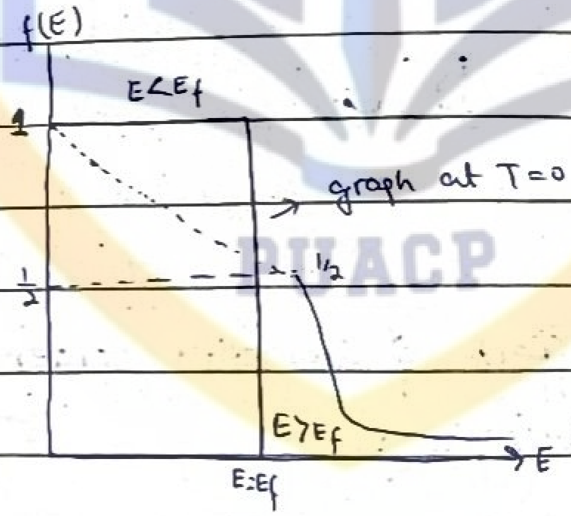
At T = 300k:-

- The step function becomes smoother.
- Electrons can occupy energy states slightly above E_f.
- Some states just below E_f are unoccupied.

At T = ∞:-

$f(E) = \frac{1}{2}$ for all E

.All energy states are equally likely to be occupied.



Question No 3

Discuss the variation in conductivity of metals with rise in temperature and impurity content?

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→ This results in more scattering of conduction electrons, causing increased resistance.

→ Hence even a small percentage of impurity can significally reduce the conductivity.

Question No: 4

Comment on conductivity when a band is filled (i) more than half (ii) completely.

Answer:-

Conductivity depends on the availability of empty energy state near the Fermi level that electrons can move into. Here's what happend in each case.

1 → **More than half-filled band:-**

- Conductive:

→ when a band is more-than half; there are still empty states within the same band.

→ Electrons can gain energy and move into these nearby emply states, allowing electrical conduction.

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★. Effect of Temperature on Conductivity:-

→. Conductivity decreases with rise in temperature.

→. In metals, free electrons are responsible for conduction.

→. As temperature increases, the lattice ions vibrate more intensely, leading to more frequent collisions between electrons and lattice ions.

→. These collisions increases the resistance, thereby reducing the conductivity.

$$\sigma \propto \frac{1}{T}$$

σ = conductivity, T = temperature

★. Effect of impurity content on conductivity:

→ impurities decreases the conductivity of metals.

→. when impurities are added, they disturb the regular arrangement of the metal atoms.

DAY:

→ This is typical of metal.

2. Completely filled band:-

- Non-conductive (insulator or semiconductor)

→ When a band is completely filled, there are no available states within the band for electron to move into.

→ For conduction to occur, electron must jump to the next higher empty band (conduction band), which a band gap.

→ If the band gap is large → insulator
if small → semiconductor.

Question No 5

How you will identify N-type and P-type semiconductor?

Answer:-

To identify N-type and P-type semiconductors.

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1. Dopent - Used:-

- N-type: Doped with pentavalent element (eg. phosphorow, arsenic).
- P-type:- Doped with trivalent elements. eg. boron, gallium.

2. Majority Charge Carriers:-

N-Type:- Electrons are majority charge carriers.

P-Type: Holes are majority charge carriers.

3. Hall effect Test:-

- Apply megnetic field and measure Hall voltage.
- Negative Hall voltage → N-Type
- Positive Hall voltage → P-Type

4. Conductivity Behavior:

N-Typ: Conductivity increase due to free electrons.

P-Type: conductivity increases due to Holes.

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5- **Thermoelectric effect (seedbeck Effect).**

if you heat one end of a semiconductor and measure the voltage difference.

- **N-Type:** Electrons move toward the cold end —— negative voltage at the hot end
- **P-Type:** Holes move toward the cold end —— Positive voltage at the hot end.

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Question 6:

Write down the properties of holes?

- $K_h = -K_e$

The wavevectors of electron band and hole are equal in magnitude but opposite in direction. If an electron of wave vector $\vec{K_e}$ is emitted from valence band then

$$\vec{K_h} = -\vec{K_e}$$

- Energy of hole is equal to energy of electron but in opposite ~~direction~~ sign. If the band is symmetric then

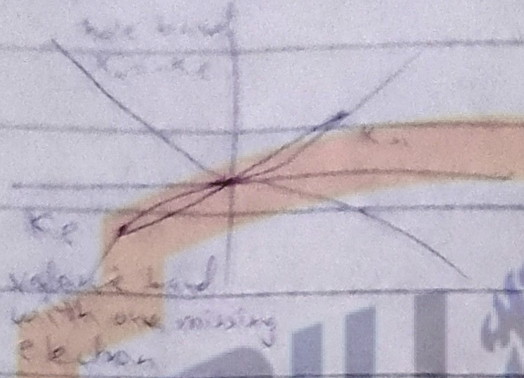
$$\epsilon_e(K_e) = \epsilon_e(-K_e) = -\epsilon_h(-K_e)$$

$$= -\epsilon_h(K_h)$$

$$\epsilon_h(K_h) = -E_e(K_e)$$

- $V_h = V_e$

The velocity of hole is equal to velocity of missing electron



Question no 7

State bloch theorem. Discuss its importance in band theory.

Bloch's theorem state that the solution of schrödinger equation for electrons in a periodic potential can be written in the form of

$\psi(r) = U(r) * \exp(ik \cdot r)$

The Bloch theorem enables us to calculate the electronics wave functions ~~and electron~~ formation of energy bands and gaps. Bloch theorem provides the mathematical framework for understanding the electronic structure of crystalline solids and their associated electrical and thermal properties

Question 8:-

Compare the results of heat capacity of electron gas calulated classical theory and quantum theory.

The heat capacity of an electron gas, as predicted by classical and quantum theories differs significantly.

Classical theory, using Maxwell Boltzmann statistics, predicts a high heat capacity, roughly 3/2 times the Boltzmann constant per electron whereas quantum theory using Fermi-Dirac Statistics, predicts a much lower heat capacity proportional to temperature and the density of states at the Fermi level.

Question 9:-

Is there any possibility of holes conduction in metals; if so justify.

No, conduction through holes is generally not a significant factor in metals. Metals primarily conduct electricity via the movement of free electrons, which occupy the valence band and conduction band. In metals the valence and conduction bands overlap, meaning electrons can easily move into higher energy levels.

How the free electron theory does explain the observed small values of specific heat of metals.

Free electron theory, using Fermi-Dirac statistics, shows that only a small fraction of electrons near the Fermi level can be thermally excited at room temperature. As a result, the electronic contribution

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to specific heat is much smaller than predicted by classical theory

Long Questions

What do you understand by fermi gas?

A Fermi gas or Free Electron gas is a collection of non-interacting fermions.

"A Fermi gas is a model in Physics describing a collection of non-interacting fermions, like electrons, protons or neutrons, that obey the Pauli Exclusion Principle."

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(12)

Give condition of crystal reflection and discuss its consequences.

The condition for crystal reflection is known as Bragg's Law, that is;

"The difference in path length of X-rays scattered by adjacent atomic planes with a crystal equals an integer multiple of the X-rays wavelength."

This condition leads to the constructive interference, resulting in a reflecting beam.

Bragg's Law is mathematically expressed as:

$$n\lambda = 2d\sin\theta$$

Consequences:

In solid state physics, crystal reflecting condition (Bragg's law) has significant consequences for understanding crystal structure and their interactions with radiation, understanding structure defects and Imperfections and to form other types of radiations.

(13)

Explain the impact of temperature on mobility(μ) of semiconductor.

The mobility means the movement of charge carriers. The mobility of an intrinsic semiconductor decreases with increase in temperature, because at high temperature, the no. of carriers are more and they are energetic also. This causes an increased no. of collisions of charge carriers with atoms and thus the mobility (μ) decreases.

(14)

Explain the principle of of Hall's effect with the help of relevant diagram.

Hall's Principle:

"The Hall effect is a phenomenon where a voltage develops across a conductor or semiconductor when it carries a current and is placed in a magnetic field perpendicular to the current."

Diagram:

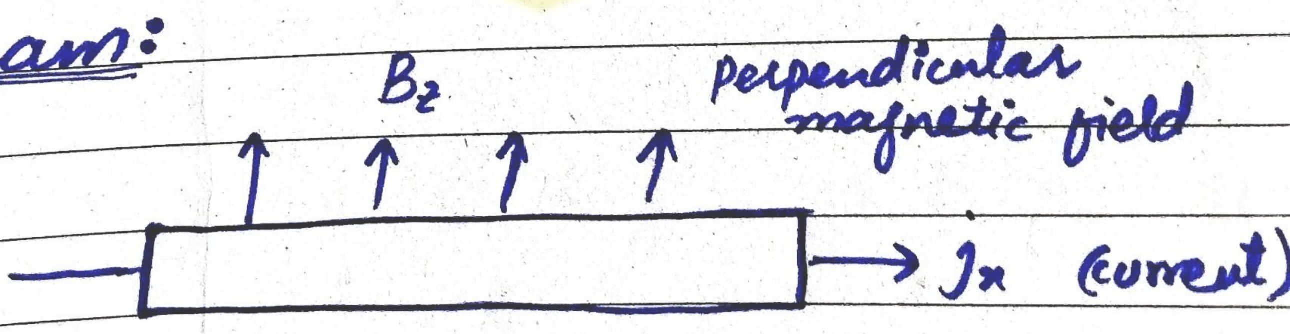

fig.(a)

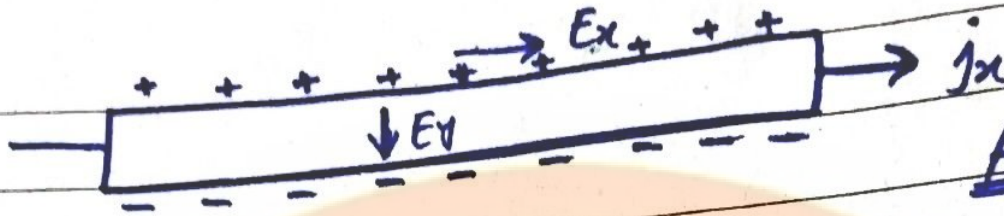

fig. (b)

(15)

Will impurities in a semiconductor effect the conductivity of semiconductor?

Ans:

Yes, impurities in a semiconductor can significantly affect the conductivity of semiconductor. The introduction of impurities into a semiconductor is called doping. Doping can either increase or decrease the conductivity of a semiconductor, depending upon the type of impurity used.

Both N-typ and P-type impurities with extra electrons (donors) and extra holes (acceptors) respectivelly enhance the conductivity of semiconductor.

Long
DATE: Q# 4 (b) part DAY:

state law of mass action. Give its significance.

Law of mass action:-

when p is greater then n will decreases as same ratio in conduction band. its product remains same. This is law of mass action.

$$np = 4\left(\frac{k_B T}{2\pi\hbar^2}\right)^3 (m_e m_h)^{3/2} e^{-E_g/k_B T}$$

n = concentration of electron

p = ~~coneto~~ concentration of holes

Where T is constant then np is constant. It is not valid only for intrinsic semi-conductor but also for extrinsic semi-conductor.

So,

in n-type when electron increases then p decreases is the same rate. So, np remains constant at any

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temperature. Similarly in p-type semi-conductor when electron decreases then p increases with the same rate so np remains constant.

Significance:-

The law of mass Action in semiconductors dictates the relationship between electron and hole concentration under thermal equilibrium, stating that their product is a constant ($np = n_i^2$), where n_i is the intrinsic carrier concentration.

→ It helps understand the behavior of charge carriers in semiconductors, regardless of doping levels and is crucial for designing and analyzing semiconductor devices.

→ This law establishes that the product of electron (n) and hole (p) concentrations is constant at a given temperature and band gap, even when the semiconductor is doped.

→ Useful for calculating minority concentration in the doped material

Long Questions

Question no 2 Pg # 197-19

Question no 3 (b)

Pg # 164 to 166 (Kittle)

Nearly free electron model

Question no3 (a)

Pg # 181 from (Kittle)

Question no 4 (a)

Pg # 153 & 154

(from Kittle)